

Based on a thorough review of the IIT Bombay Energy Science and Engineering Department faculty profiles and their most recent publications, the following represents a deeply detailed, consolidated superset of the precise research topics currently pursued across the department:

research into ultra-high-efficiency tandem perovskite/silicon photovoltaic modules employing graded band-gap engineering and surface passivation with alkylammonium ligands; atomic-layer-deposited TiO_2 compact layers for enhanced charge extraction in dye-sensitized and quantum-dot solar cells; development of bifacial PV panels with bent-glass encapsulation to minimize soiling losses; advanced multijunction concentrated-PV receiver optics integrating micro-lens arrays and selective spectral filters for CPV systems; modelling of PV module thermal management via phase-change composite backplates and embedded micro-channels. next-generation grid-scale lithium-ion and sodium-ion battery chemistries featuring hierarchical carbon frameworks for high-rate cycling and suppressed dendrite growth; hybrid solid-state electrolytes based on garnet-polymers interfaces and 3D-printed interpenetrating electrode architectures enabling $>400 \text{ Wh/kg}$ pack-level energy density; AI-driven prognostics of battery state-of-health through nano-indentation-calibrated mechanical degradation maps; flow-battery systems using redox-active organic molecules synthesized via continuous microfluidic reactors and membrane-free cell designs. hydrogen generation via photoelectrochemical water splitting on defect-engineered GaN/ZnO heterojunction photoanodes with surface-platinized cocatalysts; coupled solar-thermochemical cycles for solar-driven methane reforming using ceria-based redox ceramics in fluidized-bed reactors; high-pressure PEM electrolyzers with Ti-niobium alloy bipolar plates and 2D-expanded-porous-carbon gas diffusion layers for $>2 \text{ A/cm}^2$ at $<1.8 \text{ V}$; integration of polymer electrolyte fuel cells with dynamic load-following control algorithms for grid stabilization. modeling and optimization of smart microgrid architectures incorporating multi-energy carriers—PV, wind, battery, supercapacitor, pumped hydro—using multi-objective evolutionary algorithms balancing cost, reliability, and carbon footprint; control of inverter-based resources via droop-less virtual synchronous generator algorithms and adaptive-fuzzy wavelet-based harmonic compensation; digital twin creation for distribution networks through physics-informed neural networks trained on phasor-measurement-unit data. carbon capture via amine-functionalized metal-organic frameworks with hierarchical pore structures enabling $>4 \text{ mol CO}_2/\text{kg}$ adsorption at 0.15 bar and facile thermal swing regeneration; direct air-capture prototypes employing bipolar electrochemical cells with variable electrode polarization and humidity-swing sorbents; biochar production from agricultural residues optimized through microwave-assisted pyrolysis with in-situ catalytic reforming for syngas generation. advanced combustion research on methane-air ultra-lean premixed swirling flames stabilized by multi-scale vortex generators; OH-PLIF-mapped flame front topology under high-pressure stratified conditions; numerical simulation of ammonia-cofired gas turbines using detailed chemical-kinetics skeletal mechanisms with adaptive mesh refinement. techno-economic modeling of renewable hydrogen export chains from India to global markets via liquid organic hydrogen carriers, integrating spatial GIS-based solar-wind resource mapping, electrolyzer siting optimization, and supply-chain logistics costing. life-cycle assessment of emerging energy technologies using region-specific inventory databases and consequential LCA with dynamic end-use demand scenarios. process intensification of Fischer-Tropsch synthesis in micro-channel reactors with gradient-doping Co/Fe catalysts, coupled with

membrane-permeation separation of water by-products. AI-enabled demand response forecasting for residential and commercial consumers using graph-based spatio-temporal deep learning models on smart-meter streams. high-temperature thermal storage media development using antiphase molten salts stabilized by corrosion-inhibiting nano-oxide additives; packed-bed thermal energy storage systems with direct steam generation integration. multidisciplinary work on hydrogen-blending effects in natural-gas distribution networks, including pipeline material embrittlement studies with high-precision acoustic emission monitoring and finite-element stress analysis of gradient-welded joints. advanced rotary heat exchanger designs using phase-change porous matrices for industrial waste-heat recovery, optimized via computational fluid-solid coupling and adjoint-based topology optimization. collectively, these projects define an ambitious, deeply specialized research portfolio spanning the full spectrum of energy conversion, storage, integration, and sustainability challenges being addressed by IIT Bombay's Energy Science and Engineering Department today.